

1.Introduction and objectives

In this spatial data analysis project, we delve into the Australian Temperature dataset for the year 1980. Our primary objectives include visualizing the data, conducting exploratory data analysis, and applying various interpolation techniques. Specifically, we will perform the following tasks:

1. **Data Visualization and Exploratory Data Analysis (EDA):** We will create informative visualizations to gain insights into the temperature patterns across different regions of Australia. EDA will involve examining summary statistics, identifying outliers, and understanding the distribution of temperature values.
2. **Trend Surface Model (TSM):** To capture spatial trends in temperature, we'll employ a trend surface model. This technique allows us to estimate a smooth surface that represents the underlying temperature variation across the entire study area.
3. **Data Interpolation Techniques:**
 - **Kriging:** Kriging is a geostatistical method that predicts temperature values at unobserved locations based on nearby measurements. We'll explore kriging since we only have 1 variable.
 - **Inverse Distance Weighting (IDW):** IDW assigns weights to neighboring data points based on their distance, providing interpolated values for unsampled locations.
 - **Geographically Weighted Regression (GWR):** GWR accounts for spatial heterogeneity by fitting regression models locally at each location. We'll assess how temperature relates to other covariates across different regions.
4. **Conclusion and Recommendations:** Finally, we'll summarize our findings, highlight significant spatial patterns, and offer recommendations for policymakers, researchers, and stakeholders. Our goal is to enhance our understanding of temperature variations in Australia and inform decision-making processes.

2.Data description

The Australian Temperature dataset, sourced from Kaggle, comprises weather observations from **112 weather stations** across Australia. This rich dataset contains information on the following variables:

1. **Longitude and Latitude:** The geographical coordinates of each weather station, providing spatial context for temperature measurements.
2. **Elevation:** The elevation (altitude) of each station, which can influence local climate patterns.
3. **Year:** The year of observation, spanning from 1975 to 2019.
4. **Temperature:** The recorded temperature (in degrees Celsius) at each station.
5. **Station Name:** The unique identifier for each weather station.

Key dataset characteristics:

- **Observations:** A total of **5037 observations** are available, covering a wide temporal range.
- **Missing Values:** There are **no missing values** in this dataset, ensuring data completeness.
- **Focus on 1980:** For our analysis, we will narrow our scope to the year **1980**, which includes **112 observations** specifically for that year.

3.Data analysis

3.1 Data exploration and Visualization:

We should first explore the spatial autocorrelation in the spatial process which is the Temperature in Australia in 1980. We will use to do this the global **Moran's I** and the localized version of it, but before we conclude about them, we should check whether the spatial process follows the **Gaussian** distribution or not so that we will be able to test the significance of the above referred measures and compare the observed with the expected. We can show the test of normality in **table (3.1.1)**:

Table (3.1.1): Test the normality of the spatial process (Temperature).

Test	Variable	Statistic	P-value	Normality
Anderson-Darling	Latitude	1.2002	0.0038	No
Anderson-Darling	Longitude	5.3136	<0.001	No
Anderson-Darling	Temperature	0.3565	0.4511	<u>Yes</u>

From **table (3.1.1)**, we can conclude that the spatial process (Temperature) follows the normal distribution at significance level $\alpha = 0.05$.

Then we will use the spherical law of cosines to compute the distances because it is the fastest of the three approaches and performs relatively well for distances above 1 m and use the formula of inverse weighting $w_{ij} = \frac{1}{d_{ij}}$ which is the common formula used to compute the weights of distances so that we can compute the global Moran's I and we will show its results in **table (3.1.2)**:

Table (3.1.2): Global Moran's I.

Observed	Expected	Standard deviation	P-value
0.3662942	-0.009009009	0.02569523	0

From table (3.1.2), we can conclude that the global Moran's I is significant different from zero at significance level $\alpha = 0.05$, since we are under the normality of the spatial process, we can compare the observed **Moran's I** by the expected one, since the observed is greater than the expected then we have a positive spatial autocorrelation which means that the nearest locations tend to be similar in Temperature which also means that we could have clusters in our dataset.

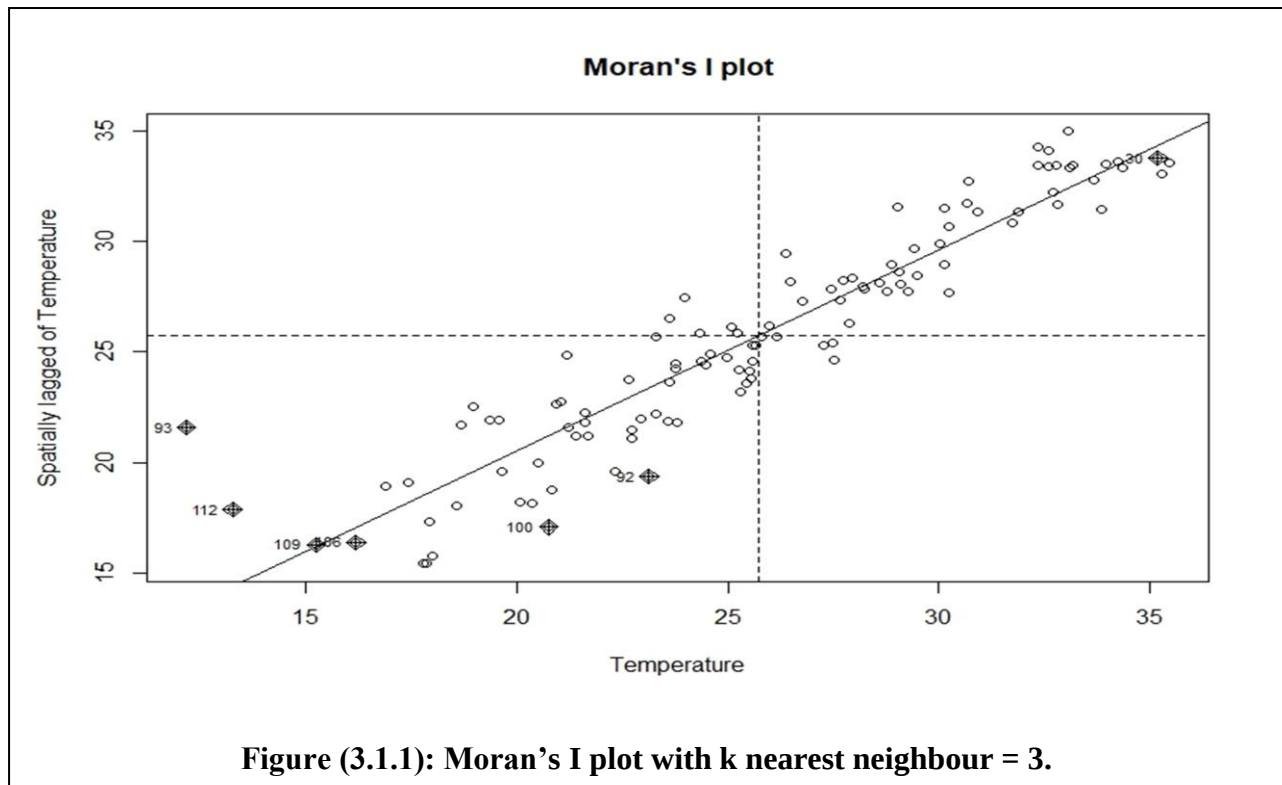


Figure (3.1.1): Moran's I plot with k nearest neighbour = 3.

From **figure (3.1.1)**, we can see from **Moran's I** plot that in general there is a positive spatial autocorrelation as the slope is positive as shown in the direction of line, we could have clusters in **High-High** area and **Low-Low** area and also may exist spatial outliers in the points which labeled with numbers above but in general we can conclude that we may not have clear outliers, we can see that in more details in the upcoming plots and initially from **figure (3.1.2)** and **figure (3.1.7)** which draws the boxplot and histogram of the spatial process (Temperature).

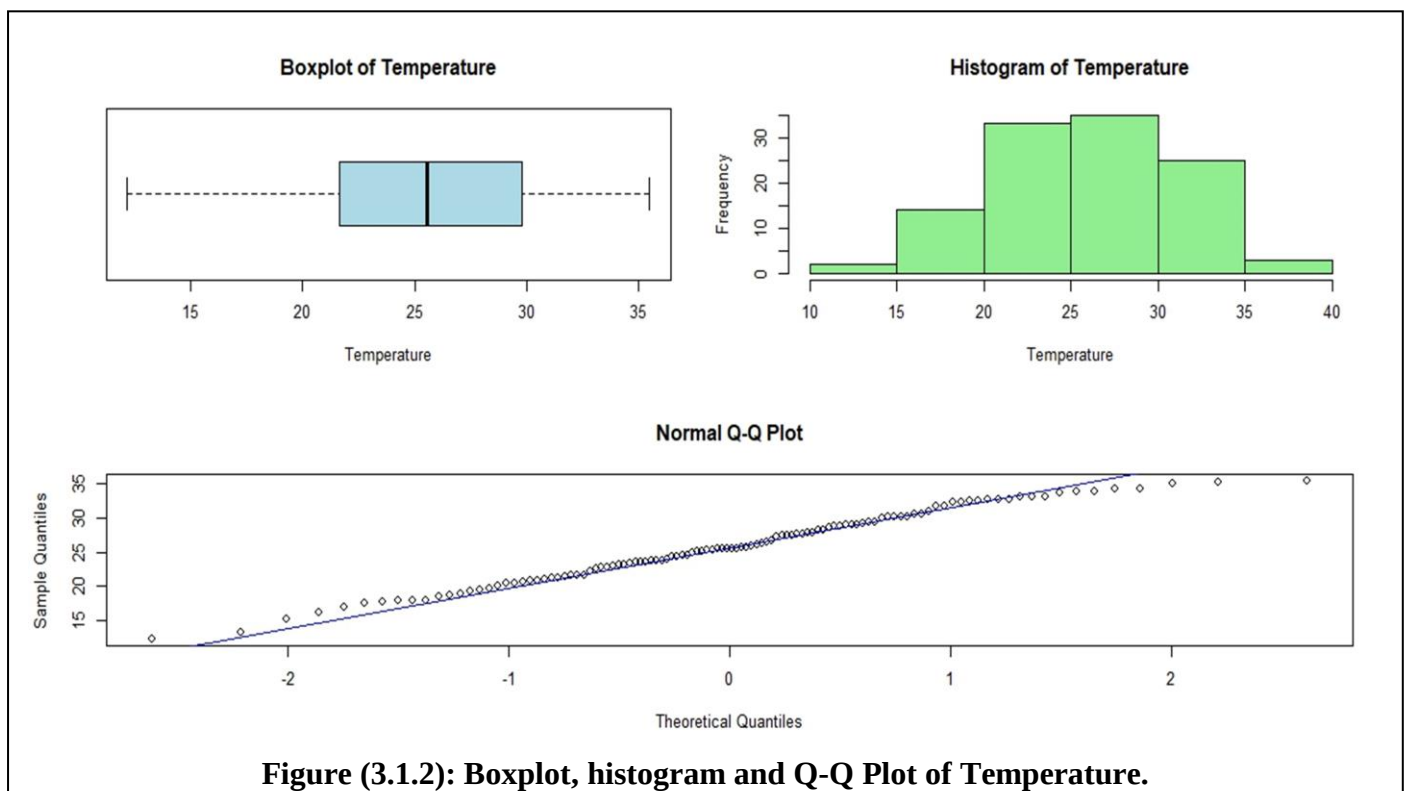


Figure (3.1.2): Boxplot, histogram and Q-Q Plot of Temperature.

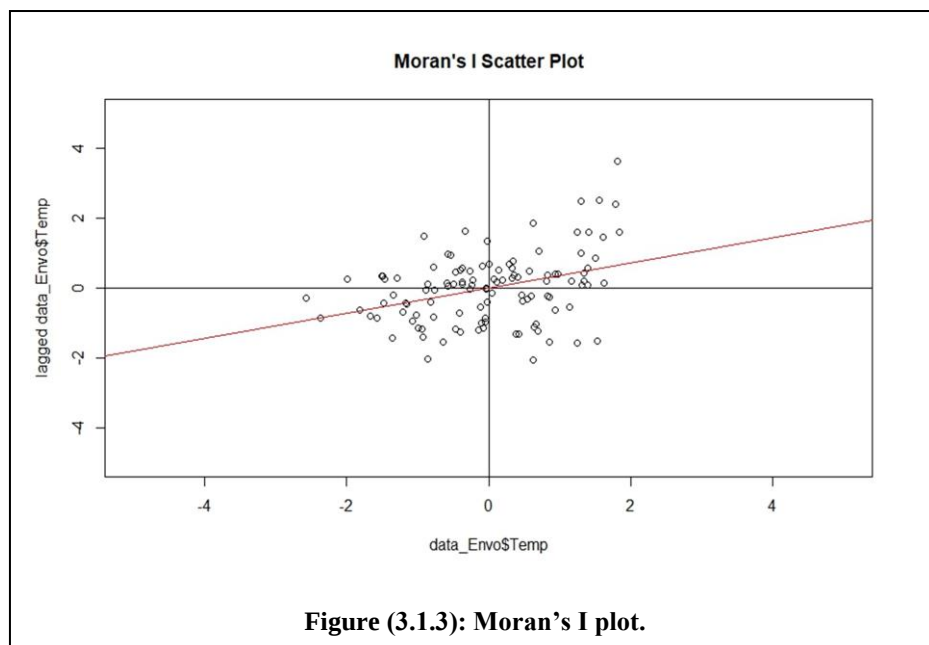
From **figure (3.1.2)**, we can conclude that there are no any outliers from the boxplot of the Temperature and also, we can see from both boxplot and histogram that the distribution of Temperature is symmetric and is bell shaped as shown in histogram, Q-Q Plot shows little deviations from the 45^o line and all of this support our test of normality in **table (3.1.1)**.

We can now explore the spatial autocorrelation at the localized level by computing the localized Moran's I in each location of the **112 locations** in our dataset by taking the number of nearest neighbors equals to six (**bandwidth = 6**), we can show that in **table (3.1.3)**

Table (3.1.3): Localized Moran's I for 15 locations out of 112 locations.

Id	I _i	E _i	V _i	Z _i	P-value
1	3.952452	-0.01595837	0.8357622	4.340854	1.419300e-05
2	3.174291	-0.01180405	0.5241298	4.400878	1.078138e-05
3	3.214531	-0.01363613	0.6302001	4.066462	4.773220e-05
4	4.997647	-0.02001053	1.5731434	4.000524	6.320247e-05
5	6.447712	-0.02502131	2.0177580	4.556728	5.195654e-06
51	1.4322563	-0.01828569	1.0969266	1.3849736	0.1660605
52	0.9430743	-0.01600587	1.0012467	0.9584829	0.3378193
53	0.1078029	-0.01218787	0.5870281	0.1566096	0.8755526
54	0.8572427	-0.01222011	0.8055040	0.9687622	0.3326638
55	1.3911496	-0.01238070	0.8948966	1.4836634	0.1378983
108	4.727276	-0.01677678	0.9705302	4.815540	1.468021e-06
109	7.966048	-0.02174013	1.7905637	5.969413	2.381085e-09
110	7.238930	-0.02363332	1.9371132	5.218101	1.807672e-07
111	7.342611	-0.02393092	2.0272533	5.173800	2.293805e-07
112	5.198034	-0.01273844	0.5305482	7.153848	8.437849e-13

From **table (3.1.3)**, we can conclude that since the spatial process (temperature) is normally distributed and by looking at p-values, **localized Moran's I** and expected of them, then we can see that there is a positive spatial autocorrelation at locations from **1** to **5** and from **108** to **112** and this means that these locations are similar to their neighbors and could exist cluster around them but there is insignificant spatial auto correlation for the locations from **51** to **55**.



From **figure (3.1.3)**, we can conclude that there is a positive spatial autocorrelation among the points which means that the closest points tend to be similar and this is shown in the slope of the red line.

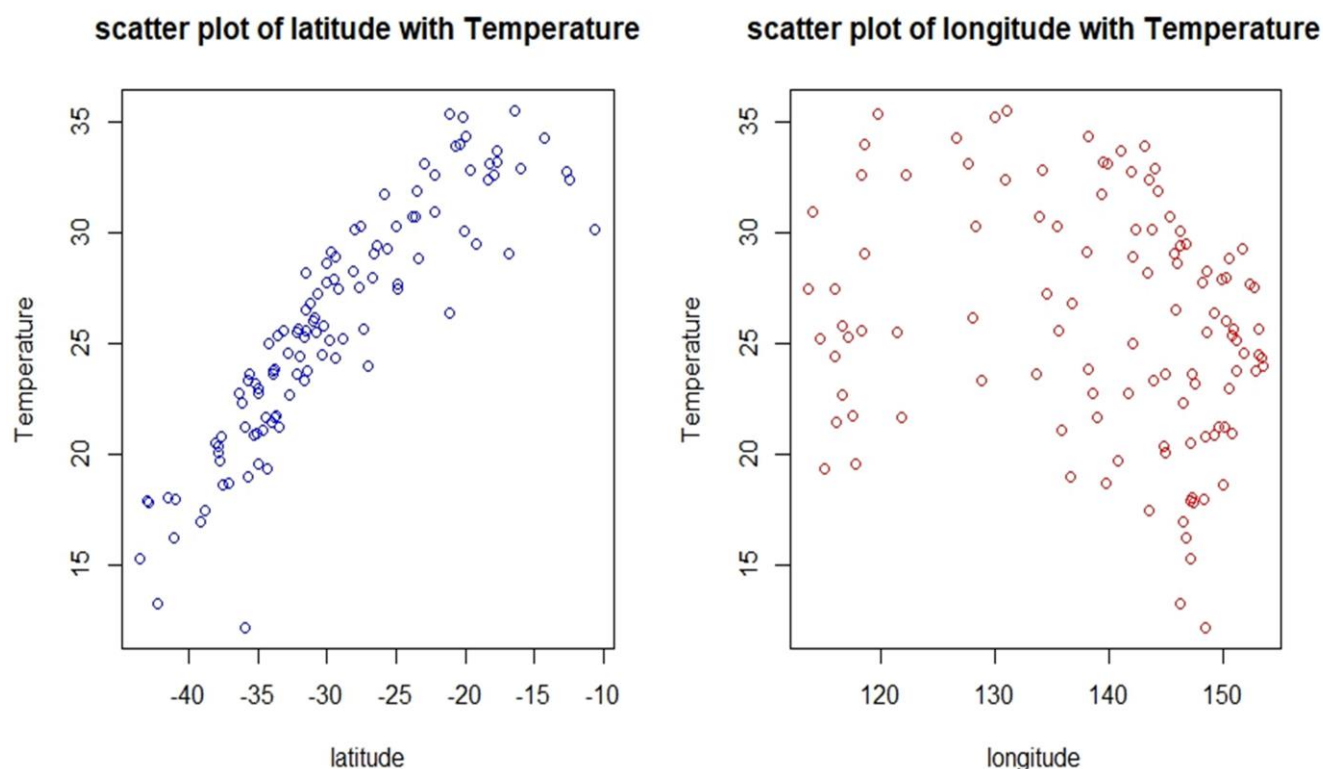


Figure (3.1.4): scatter plot of Temperature with both latitude and longitude.

From **figure (3.1.4)**, we can see that when we go or direct to the **South**, the Temperature in the area under study decreases in spite of we direct to the **North**, the Temperature increases gradually. In the other hand, when we direct to the **East**, the Temperature tends to be Moderate in spite of we direct to the **West**, the Temperature begins to increase gradually.

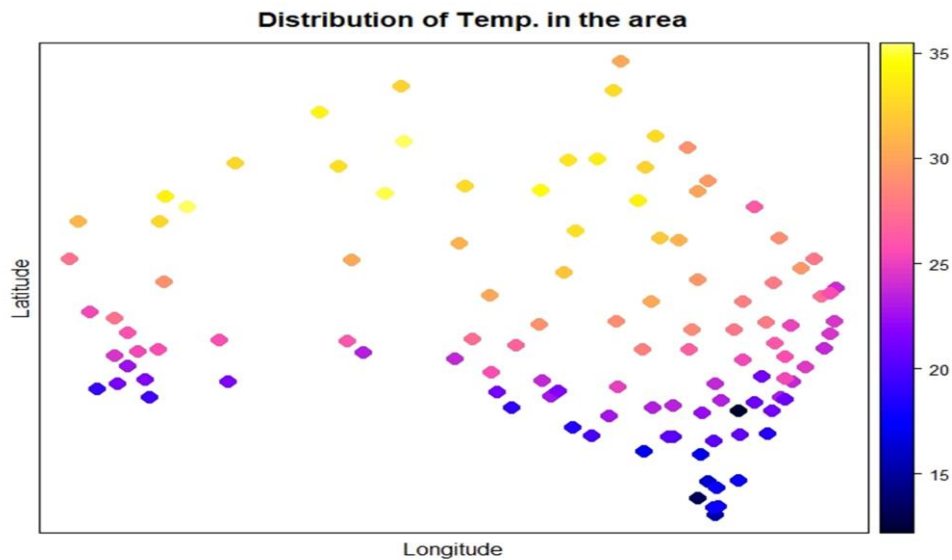


Figure (3.1.5): Sp-plot of Temperature with both longitude and latitude.

From **figure (3.1.5)**, we can see that the more we go to **North**, the highest Temperature is in spite of the more we go to **South**, the lowest Temperature is. In the other hand, the more we go to **East**, the more Temperature is moderate in spite of the more we go to **West**. Also, we can conclude that there is a cluster of the locations with high Temperature (**hot spot**) in the North in spite of there could be a cluster in the **Southeast** in the locations with low Temperature (**cold spot**), and there may exist one outlier the **black** point with lowest Temperature around the **blue** and **purple** points with moderate Temperature.

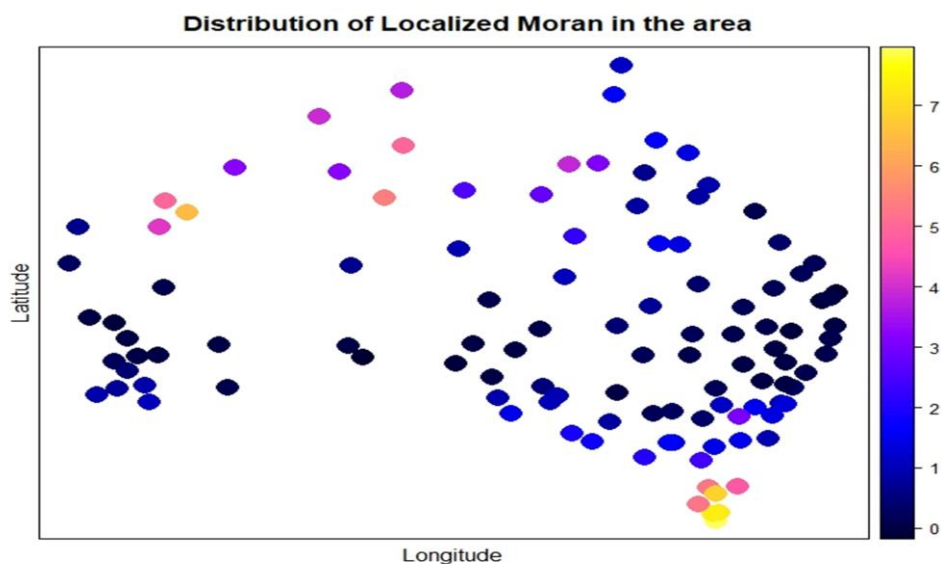
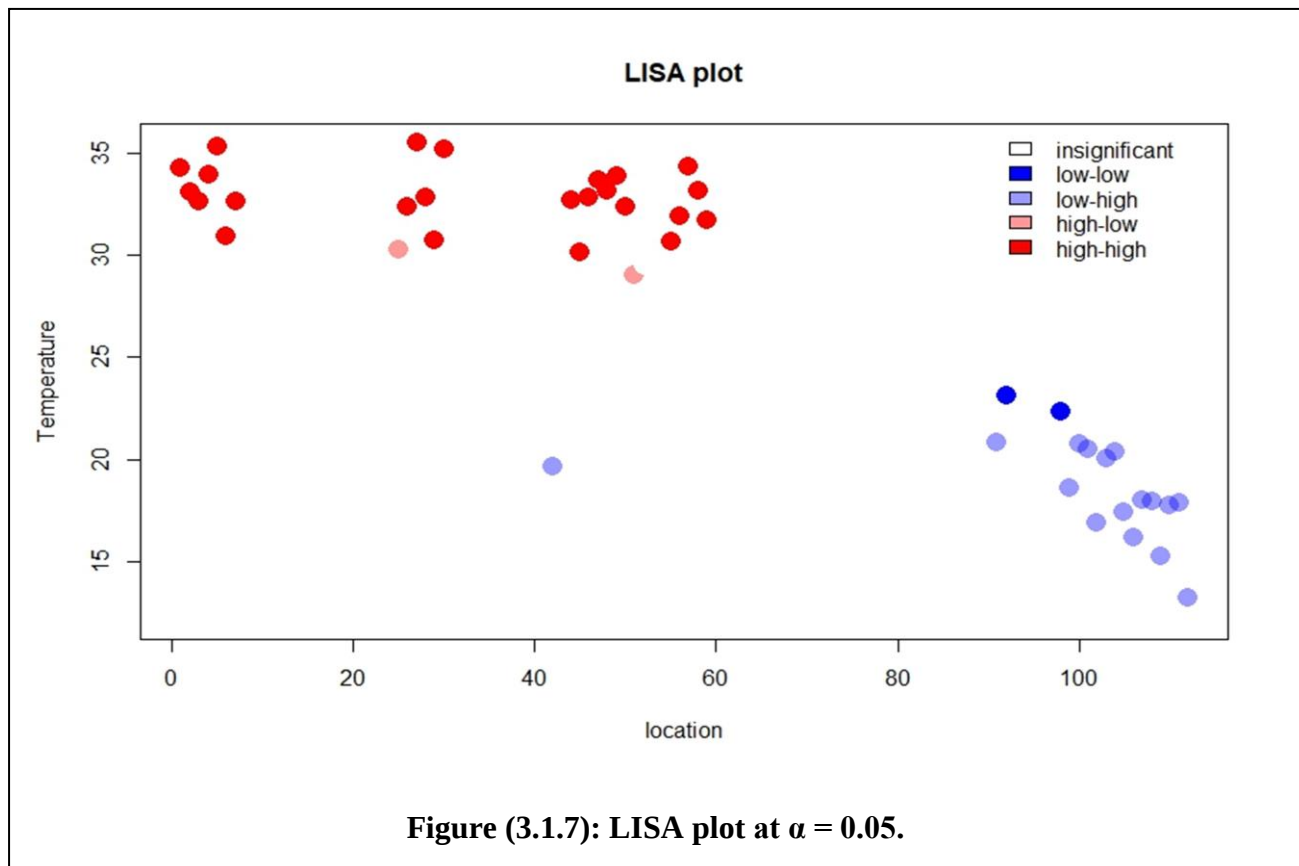


Figure (3.1.6): Sp-plot of Localized Moran's I in each location.

From **figure (3.1.6)**, we can conclude that the values of localized Moran's I increase when we direct to the **South**, and there could be cluster in the **East** with **black** points.



From **figure (3.1.7)**, we can see that there are two clusters in the data. One is considered to be High-High (**hot spot**) with the highest Temperatures and the other is considered to be Low-Low (**cold spot**) with the lowest Temperatures, all the points are significant with significance level $\alpha = 0.05$. There may exist one spatial outlier which is an alone point in light-blue around insignificant points.

3.2 Trend Surface Model (TSM):

In this model we try to predict the response variables using the coordinates of a location (longitude and latitude) through estimating the parameters in the model below:

$$Z(s_i) = x(s_i)\beta + u_i$$
$$x(s_i) = \sum_{k=0}^p \sum_{m=0}^p \beta_{km} x_i^k y_i^m \quad k + m \leq p$$

Since The degree (p) determines the complexity of model and the curves to capture the data points, we will try possible values for p which are 1, 2, 3 and 4. To obtain the best model of the 4 we will observe metrics such as Akaike information criteria AIC and Adjusted R-squared.

Metric	1st Order	2nd Order	3rd Order	4th Order
Adj.R2	0.8083	0.8926	0.908	0.9162
AIC	190.7605	128.7632	115.113	109.0795

3rd order is the best one among the others since it has relatively high adj.R2 and low AIC, Although the 4th order has the highest adj.R2 and lowest AIC but the difference between 3rd and 4th order isn't significant, therefore , to make a decision we have to consider the number of parameters to be estimated in both models, since it will affect the precision of the prediction, where 3rd order has lower number of parameters compared to 4th order, hence, we will choose the 3rd order for exploring the pattern of temperature in the Australian areas.

3D plot when $p = 3$

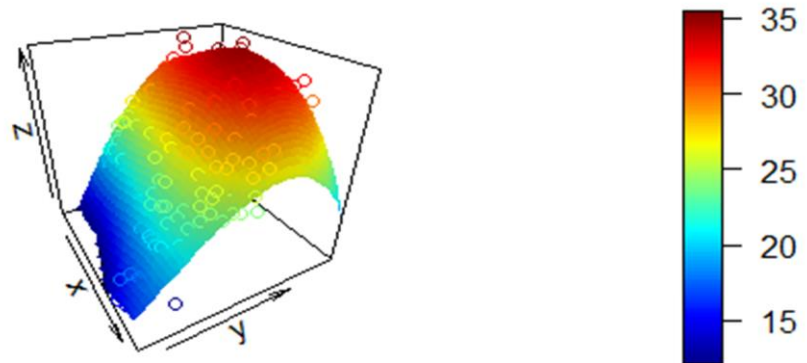


Figure (3.2.1): 3D plot for Trend surface model of order 3.

The figure above shows the 3-dimensional plot where it shows the cubic shape of the relationship between the longitude, latitude, and temperature. We will also observe the contour plot since it's easier to conclude information from.

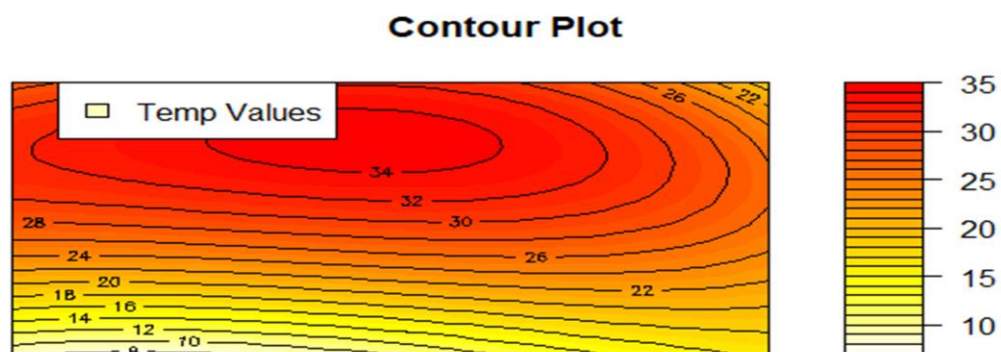
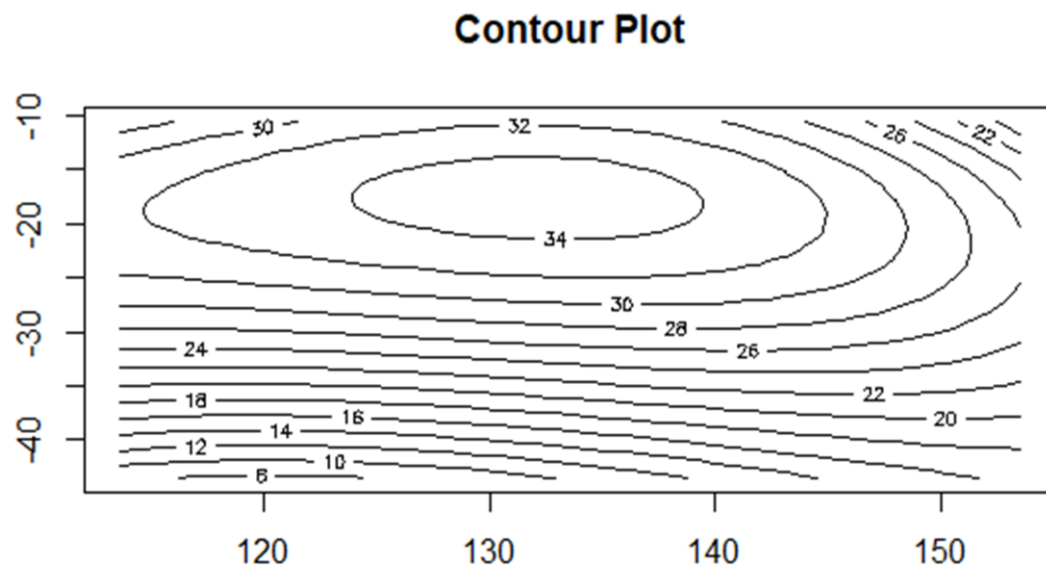
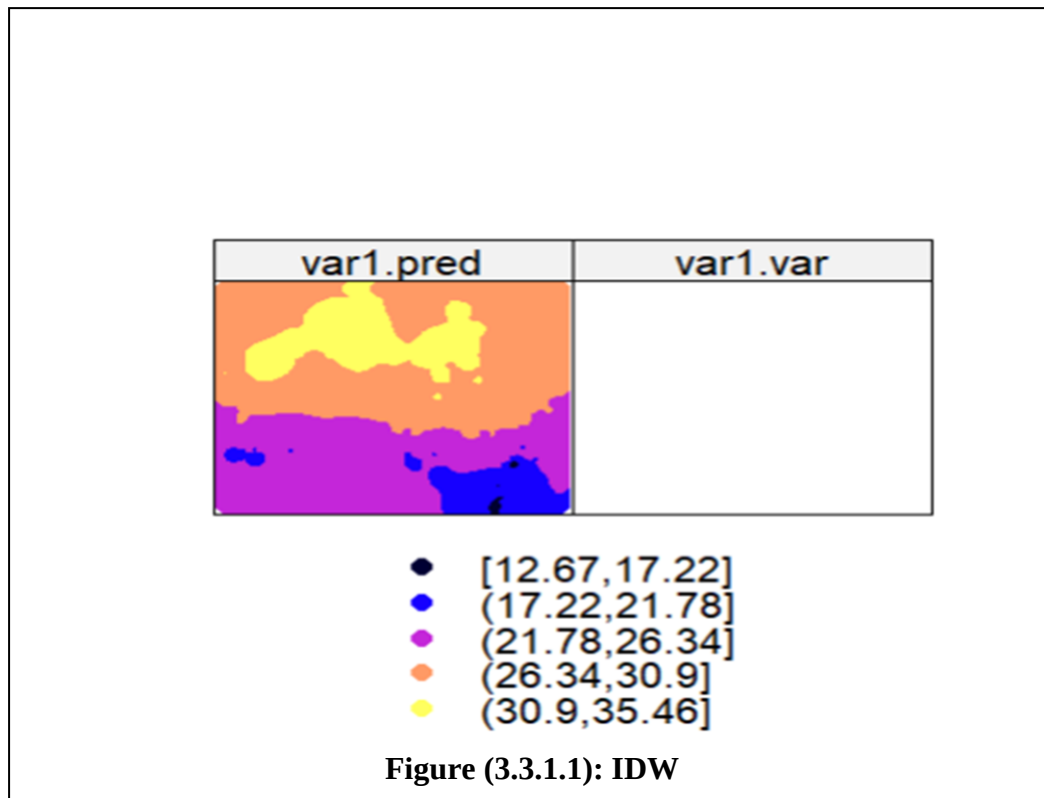


Figure (3.2.2): Contour plot for Trend surface model of order 3

The 2 plots above show the temperature values in the area under study, we can see that the highest temperature occurs at longitude 130 and latitude -20 while the coolest area is when the longitude = 120 and latitude = -40.

3.3 Spatial Interpolation:

3.3.1 Inverse Distance Weighting (IDW):

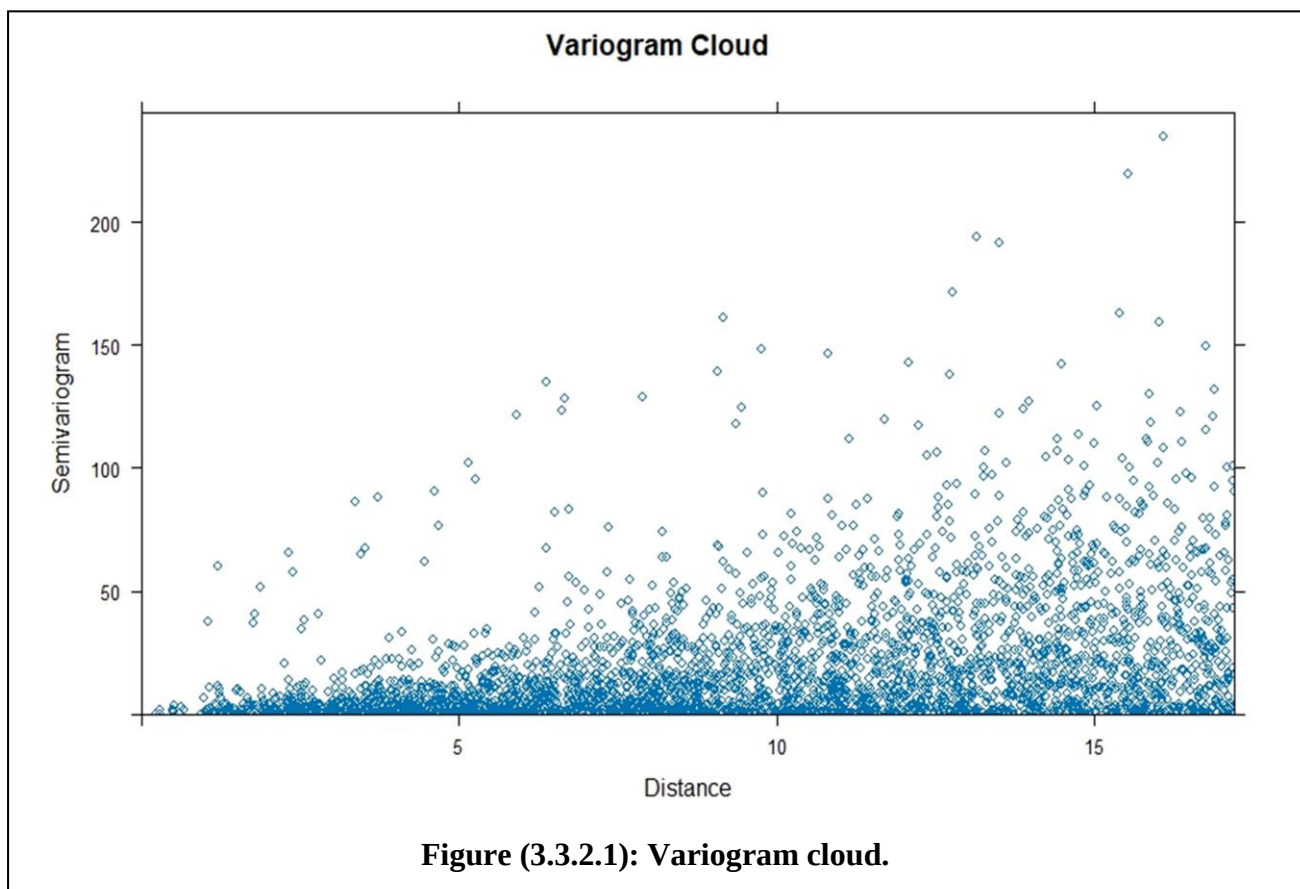


From this Figure, we find that, there is an empty plot for the variance of the interpolated points because IDW isn't a 'model' like kriging meaning that it's just a mathematical relation that doesn't have a deterministic part nor random part compared to the kriging models.

The left figure expresses the prediction for 30000 points where we can see those locations in southeast part, which are colored in black and blue, have low temperature ranging from 12.67 to 21.78, Unlike the north since as we move to the northern part the temperature increases where the locations have values ranging from 26.34 and 35.46, locations in the center of the study area tend to be moderate as the temperature ranges from 21.78 and 26.34.

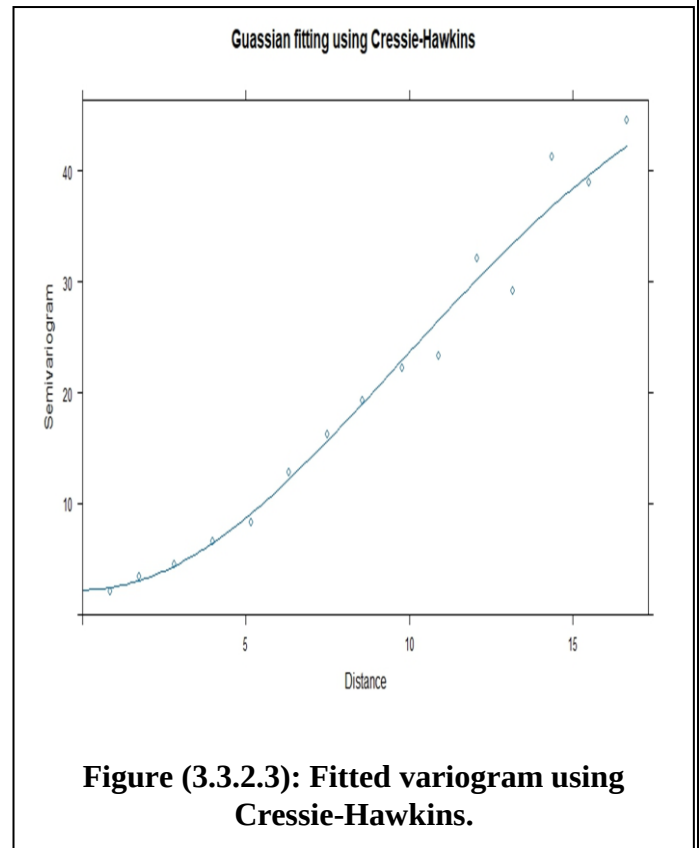
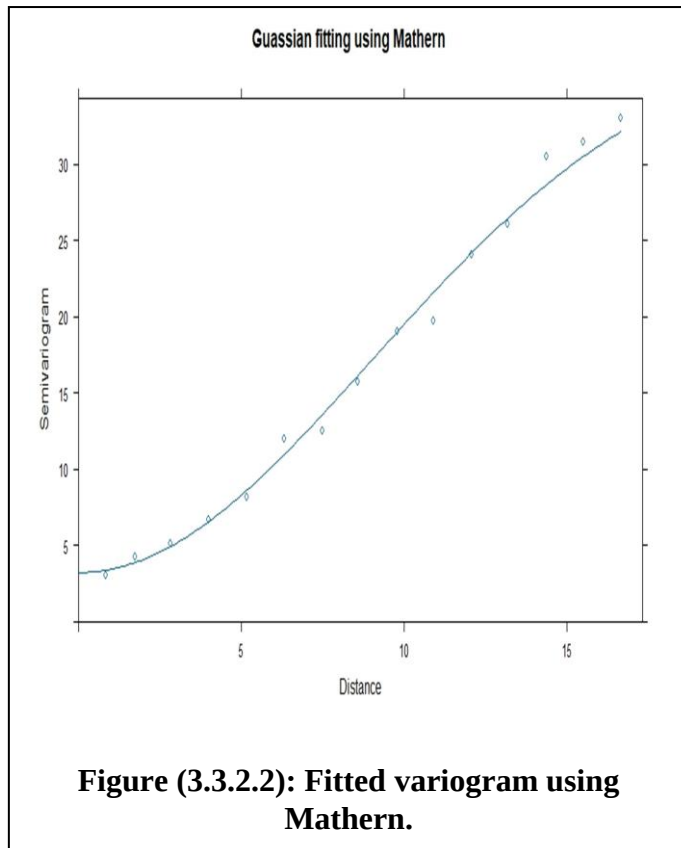
3.3.2 Kriging:

Before we are talking about the Kriging, we should compute the distances and weights to investigate the variogram and decide which variogram parametric model is the best one to fit our dataset. We first can investigate the variogram by plotting the variogram cloud and this can be shown in **figure (3.3.2.1)**.



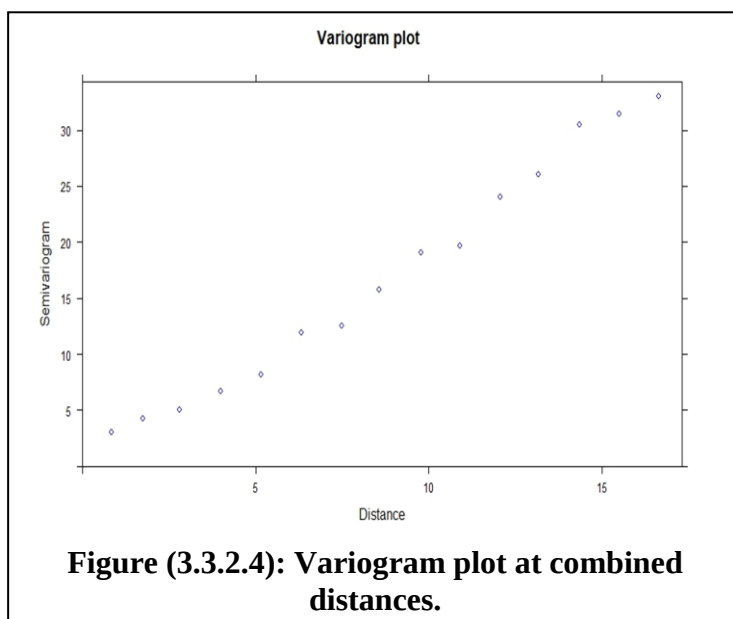
From **figure (3.3.2.1)**, we can see the values of the semi variogram increase with the increase of the distance, this plot has been drawn with the pairs of the distances not the combined distance.

Since from **figure (3.1.7)** which shows that we may have outliers, then we will use both Mathern and Cressie-Hukins in estimating the variogram to detect the differences. This can be shown in **figure (3.3.2.2)** and **figure (3.3.2.3)**.



From **figure (3.3.2.2)** and **figure (3.3.2.3)**, there is no a difference whether using Cressie in estimate the variogram or not and the fitted models using both does not different too much. both give us the best model to fit the variogram is the Gaussian but the SSE of Gaussian using Mathern = **39.6** which is less than Cressie which equals to **116.16**. so, we will complete with Mathern estimation.

Now, we will plot the variogram at the combined distance to decide which parametric model fits the data well, this can be shown in **figure (3.3.2.4)**.



From **figure (3.3.2.4)**, we can see that the variogram plot is near to both Gaussian and linear parametric model. To avoid this confusion, we will use the attribute of the **error sum of squares**, then we will choose the best fitting model which has the minimum error sum of squares.

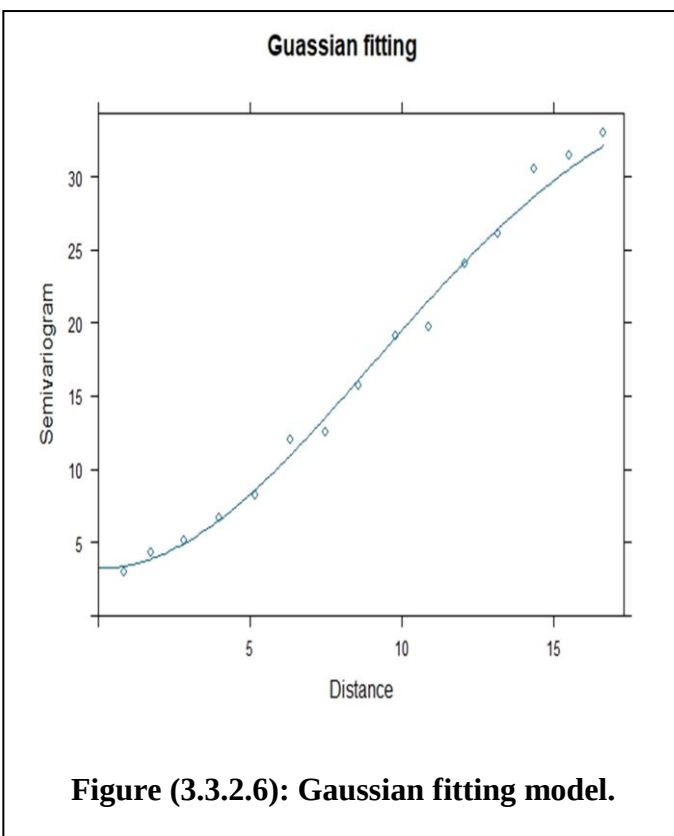
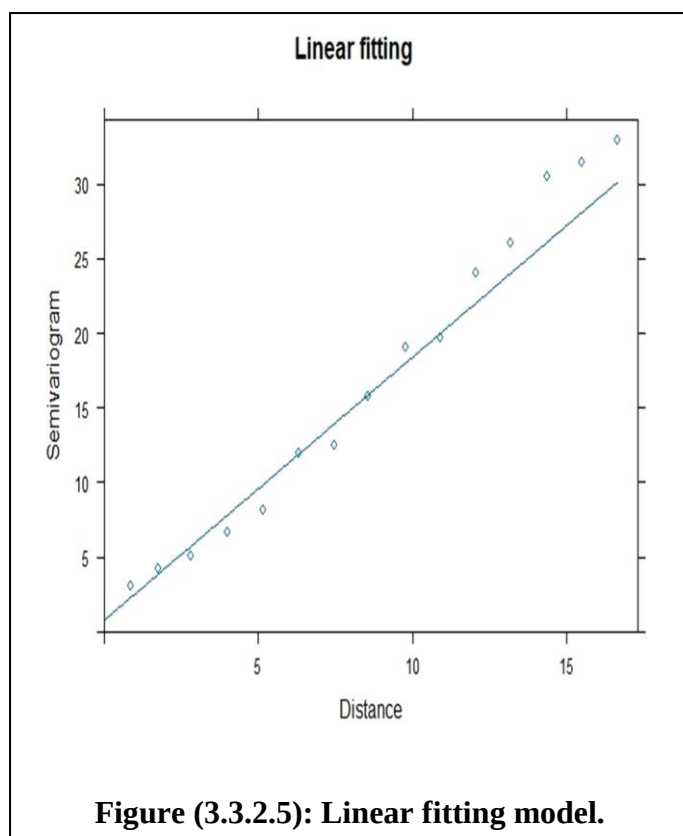


Figure (3.3.2.5): Linear fitting model.

Figure (3.3.2.6): Gaussian fitting model.

From both figure (3.3.2.5) and figure (3.3.2.6), we can see that the Gaussian and linear models are the best fitting for the variogram from any other parametric models, but it is clear that the Gaussian is more accurate than the linear model that can be illustrated in **table (3.3.2.1)** which compute the error sum of squares for each model. We have tried all possible by changing the boundaries of the variogram plot but the best (minimum) SSE exists in **table (3.3.2.1)**.

Table (3.3.2.1): The error sum of squares of the parametric models.

Parametric model	Error sum of squares
Gaussian	<u>39.76464</u>
Linear	153.9631
spherical	326.0821
Exponential	964.148

From **table (3.3.2.1)**, we can conclude that the best fitting model based on the error sum of squares is the Gaussian model with the minimum error sum of squares which is in the **underlined format**. The Gaussian variogram model can be written as follows:

$$\hat{\gamma}(h) = 3.226067 + 35.204635(1 - \exp\left(-3\frac{h^2}{12.69282^2}\right))$$

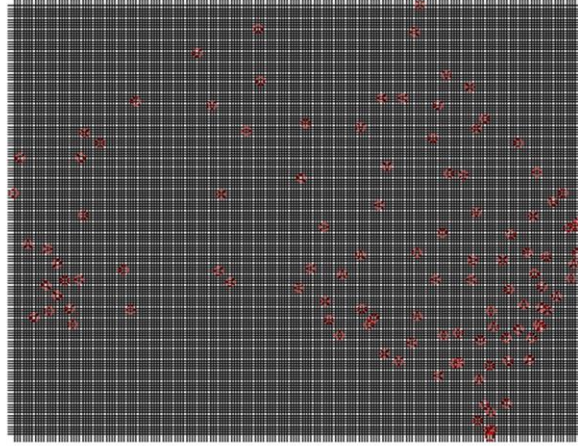


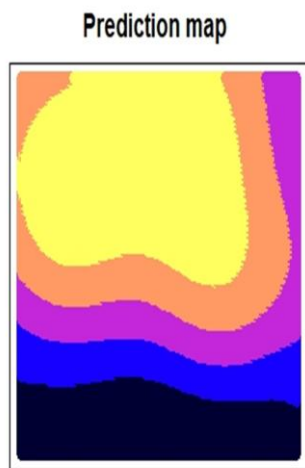
Figure (3.3.2.7): Grid.

Figure (3.3.2.7) shows the grid for the locations which contains **30000** of points to cover the whole area in an accurate way, the sample points are in the **red color**.

We have tried grid with different number of points such as 5000, 10000 and 30000, we have noticed that there is no a significant difference among them, so we have decided to use 30000 points to decrease the standard error.

Since we do not have any explanatory variables which affect the temperature and by assuming that the trend is constant but is unknown, so we have used the ordinary kriging model to predict the spatial process (Temperature) under the 2nd order stationarity to ease the complicated computations and to get the variogram and co-variogram matrix.

$$z_{ok}^*(So) = \hat{\mu} + v' \Sigma^{-1}(Z(s) - \hat{\mu}1)$$



◆ [15.13,18.97]
 ◆ [18.97,22.82]
 ◆ [22.82,26.67]
 ◆ [26.67,30.52]
 ◆ [30.52,34.37]

Figure (3.3.2.8): Prediction map.

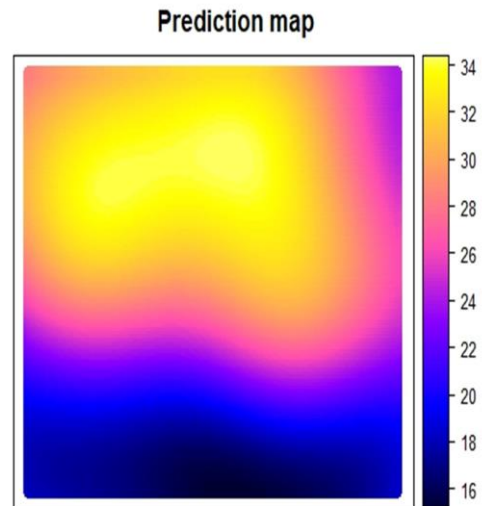
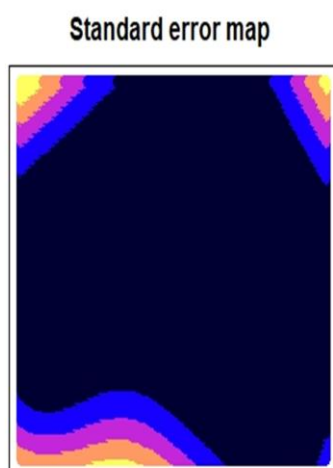


Figure (3.3.2.9): Prediction map with colorskey.

From both **figure (3.3.2.8)** and **figure (3.3.2.9)**, we can see that in general the map has divided into different layers, Temperatures increase in the North of the area in spite of the South of the area. We can also conclude that we have different clusters maybe 5 clusters, but in those clusters, we can detect that the cluster in **Yellow** is the **hotspot** cluster with the highest Temperatures. In the other hand, the cluster in **dark blue or black** is the **cold spot** cluster with the lowest Temperatures.



◆ [3.408,8.138]
 ◆ [8.138,12.87]
 ◆ [12.87,17.6]
 ◆ [17.6,22.33]
 ◆ [22.33,27.06]

Figure (3.3.2.10): Standard error map.

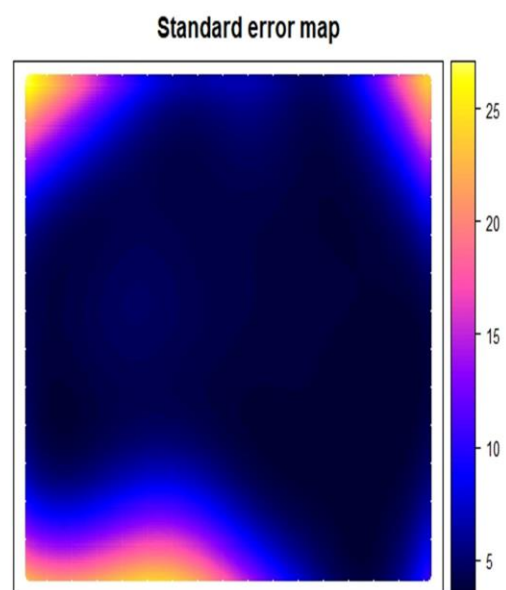


Figure (3.3.2.11): Standard error map with colorskey.

From **figure (3.3.2.10)** and **figure (3.3.2.11)**, we can see that the standard error is very low in the middle of the area in spite of the boundaries which need more and more stations in these areas as the standard error in them is very high.

3.3.3 Geographical Weighted Regression Model (GWRM):

Table (3.3.3.1): GWRM output.

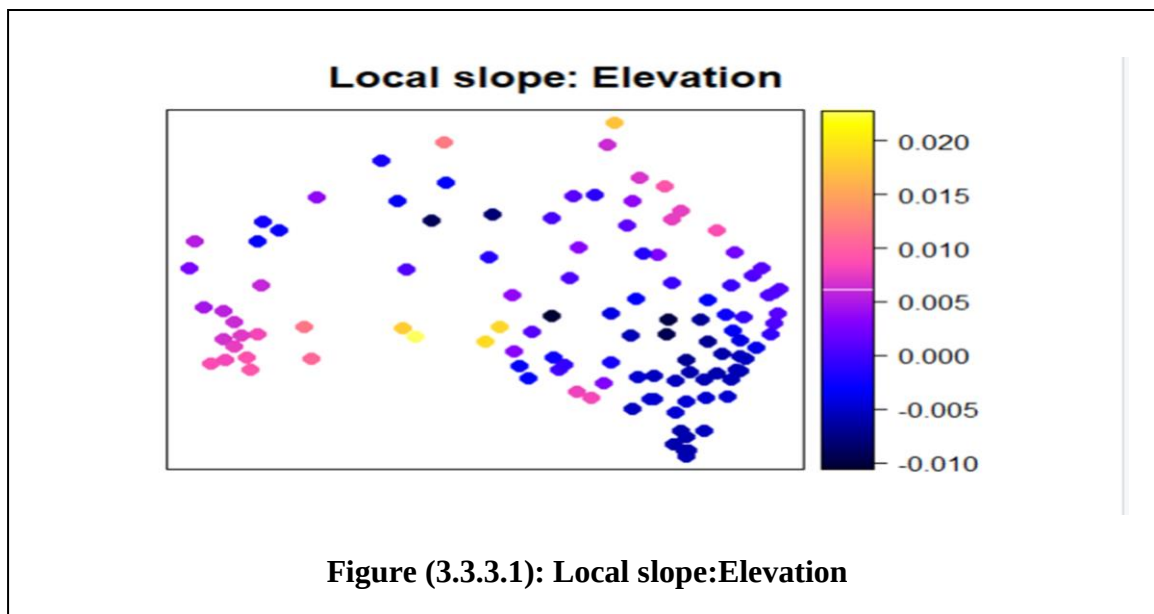
Gwr(formula=Temp~elev,data=data_envo,coords=cbind(long,lat),bandwidth=col.bw, Gweight=gwr.Gauss,hatmatrix=TRUE)					
	Min	1 st QU	2 nd Qu (median)	3 rd Qu	Max
X.intercept	17.51522601	22.30006449	24.97558782	29.28402086	37.40603445
elev	-0.01065967	-0.00503659	-0.00081637	0.00470343	0.02282311
	Global				
X.intercept	26.0511				
elev	-0.0019		Number of points:112		

After we compared between fixed and adaptive bandwidth, and compared between the type of kernel function, we use fixed bandwidth by value 2.190562, and the kernel function used is gaussian, these given us the best result.

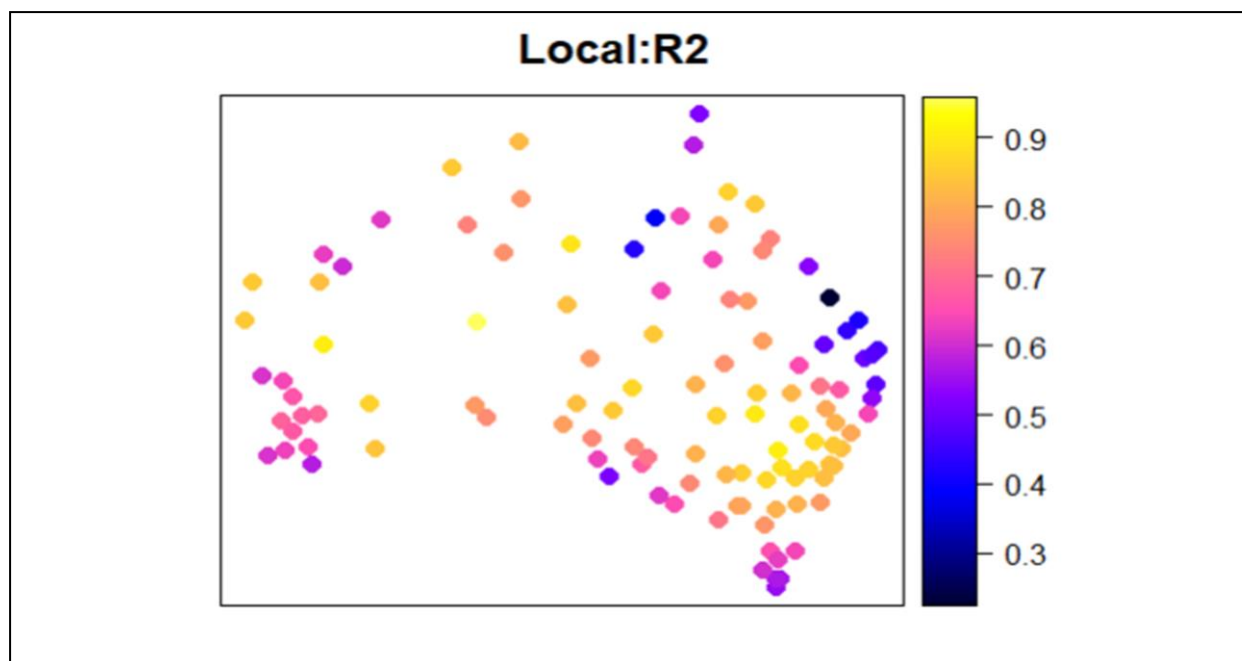
From this Table, we find that, there is a weak local variation, because there is a region, the nature of the relationship between temperature and elevation is inverse and another region the relationship is positive and there is a region it has no effect.

The value of the global model, it is the result of the presence of negative values and other positive values, on average it doesn't represent all of the data.

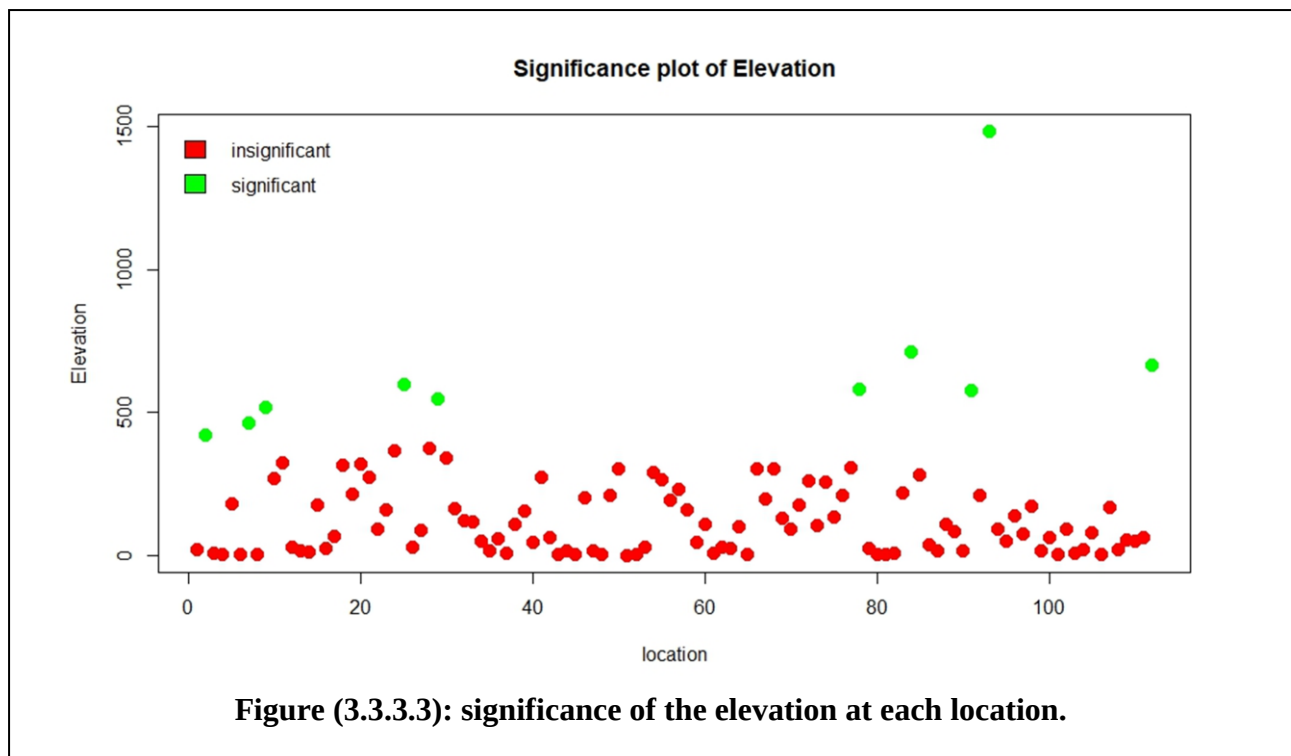
Plot for local slope: Elevation: -



From This **figure (3.3.3.1)**, we find that, in most regions, the effect of elevation is negative and weak, except for the border areas in the northeast and southwest, the effect is positive. Therefore it is varies depending on location, and this indicates that it is better to use GWRM.



From **figure (3.3.3.2)**, we find that, in general, the value of R square is high in most areas, except for some border areas in the northeast.



From **figure (3.3.3.3)**, we can see that only ten locations have significant elevation in spite of the other locations and also, the more elevation increases, the more the points tend to be significant.

4. Conclusions and recommendations

After using IDW and kriging to predict the temperature using the known locations we can conclude from the figures above that moving towards the northern part of the area under study we tend to have high temperature values, moderate weather in middle and cool weather in the southern part.

However, the temperature prediction for the outskirts of the study isn't accurate which can be seen in **Figure (3.3.2.10)** where the error tends to increase as we move to the boundaries of the study area Unlike the middle part which has the lowest error among all other locations.

Moreover, in **Figure (3.3.3.2)** we see low R-squared in the eastern part of the area under study which supports what we said above, Hence, we would advise the decision maker to build more stations on the border areas to have better accuracy when predicting the temperature while there's no need to build any stations in the middle of the area under study since we can perform the prediction with high accuracy.